

Motivation

Trade of indivisible goods between unit-demand agents with money.
Introduced by Quinzii (1984), Gale (1984), and Svensson (1984).



- House allocation problem without money solved by Gale's Top Trading Cycle algorithm.
- Housing market with **quasilinear utilities** (in money) solved, e.g., by Hungarian algorithm.

For high-value items (e.g., houses):

- agents experience income effects,
- willingness to pay depends on level of wealth.

Nothing known about computational complexity of 'more realistic' assumptions, e.g. 'soft' budget constraints or costs of borrowing.

Gale's Housing Market

- Agents $\{1, \dots, n\}$ and houses $\{1, \dots, n\}$.
- Agent i has *preference sets* $P_0^i, P_1^i, \dots, P_n^i$ covering $\mathbb{R}^n$.
- Demands house j at prices $\mathbf{p} \in P_j^i$, and **nothing** at $\mathbf{p} \in P_0^i$.
- Indifferent between houses j and k if $\mathbf{p} \in P_j^i \cap P_k^i$.

competitive equilibrium: Prices $\mathbf{p} \in \mathbb{R}^n$ and envy-free allocation $\pi : [n] \rightarrow [n]$ of houses to agents: $\mathbf{p} \in P_{\pi(i)}^i$.

ε -**approximate competitive equilibrium:** Prices $\mathbf{p} \in \mathbb{R}^n$ and allocation $\pi : [n] \rightarrow [n]$ of houses to agents s.t. $\mathbf{p}$ is ε -close to $P_{\pi(i)}^i$.

Gale's assumptions

- P_j^i are closed.
- For each house j : $\mathbf{p} \notin P_j^i$ if $p_j \geq 1$.
- $P_1^i, \dots, P_n^i$ cover lower faces of n -dimensional unit cube.

Theorem: (Gale, 1984) Competitive equilibrium exists if agents satisfy Gale's assumptions.

Theorem: ε -approximate competitive equilibrium exists if agents satisfy Gale's assumptions (ii) and (iii).

Computational Questions

How hard is it to compute (approximate) competitive equilibria?

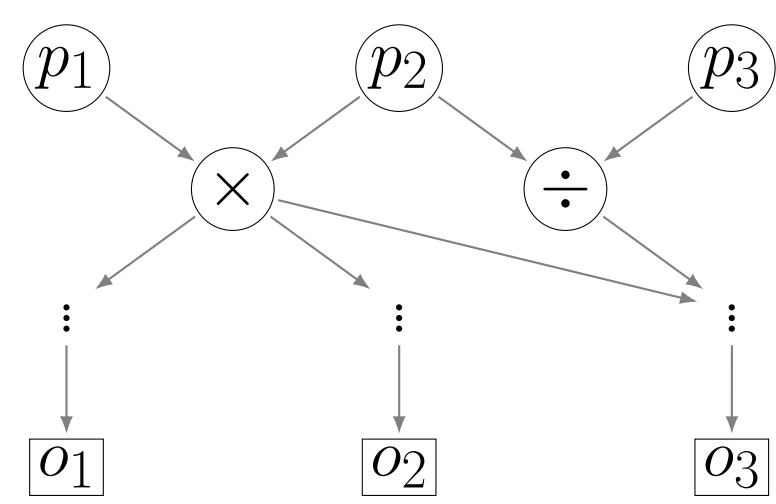
Housing: Given agent preferences that satisfy Gale's assumptions, compute equilibrium $(\mathbf{p}, \pi)$.

ε -Housing: Given $\varepsilon > 0$ and agent preferences that satisfy Gale's assumptions (ii) and (iii), compute ε -equilibrium $(\mathbf{p}, \pi)$.

Computational representation of preference sets: $P_0, P_1, \dots, P_n$

Arithmetic circuits

- Input: prices $\mathbf{p} \in \mathbb{R}^n$.
- Output: $\mathbf{o} \in \mathbb{R}^{n+1}$ with $o_j = 0$ iff $\mathbf{p} \in P_j$.
- Base $\{+, -, \times, \div, \min, \max\}$, rational constants.



Polynomial-time algorithms

- Input: prices $\mathbf{p} \in \mathbb{R}^n$.
- Output: Houses demanded at $\mathbf{p}$.
- Polynomial runtime guarantees.

```
26 function unit_demand(p: Dict{Int, Int})
27   max_utility = 0
28   Ψ = Set{Int}()
29   for u ∈ intersect(keys(p), trades)
30     u = util(u, Set{u})
31     if u > max_utility
32       max_utility = u
33       Ψ = Set{u}
34   end
35   end
36   return Ψ
```

Function oracles $f_1, \dots, f_n$.

- Input: prices $\mathbf{p} \in \mathbb{R}^n$.
- Output: $f_j(\mathbf{p}) = 1$ iff $\mathbf{p} \in P_j$.
- Similar to demand queries in auctions



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Illustration with Two Houses

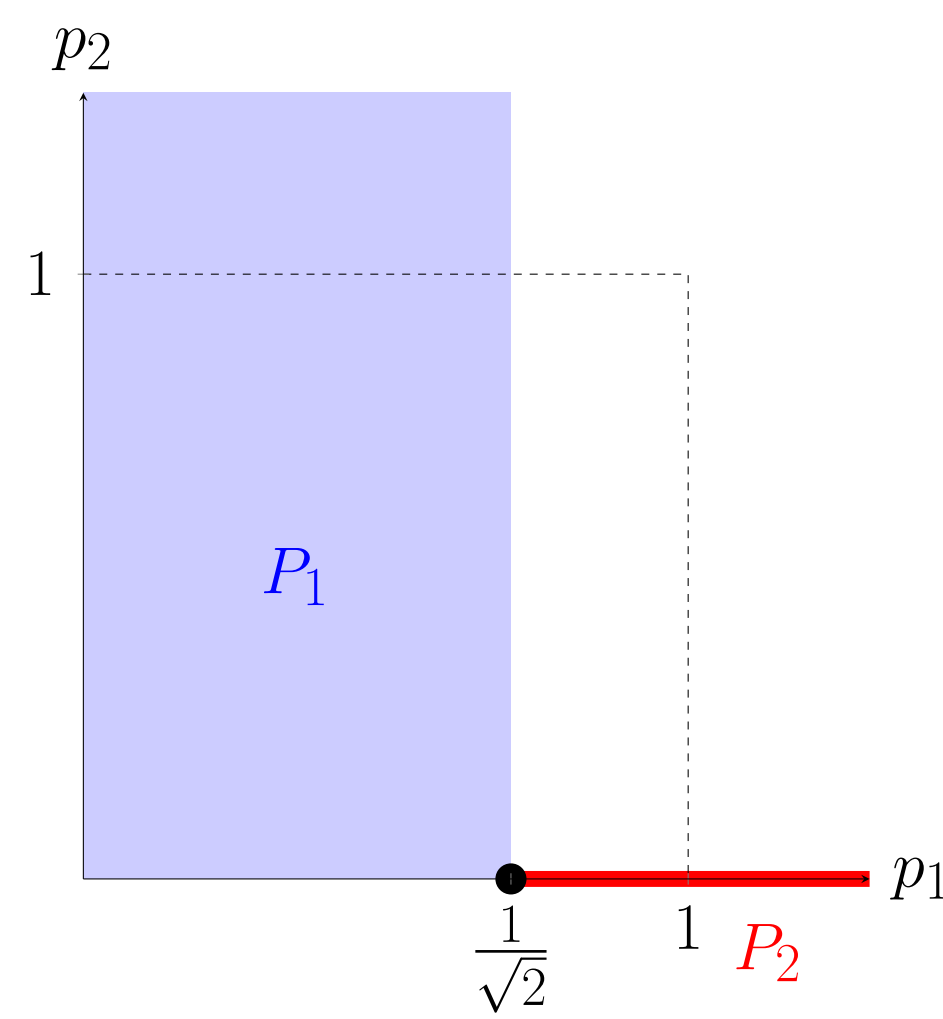
Two agents with identical preferences

$P_0 = \mathbb{R}^n$.

$P_1 = \{\mathbf{p} \in \mathbb{R}^n \mid 0 \leq p_1 \leq \frac{1}{\sqrt{2}}\}$.

$P_2 = \{\mathbf{p} \in \mathbb{R}^n \mid p_1 \geq \frac{1}{\sqrt{2}} \text{ and } p_2 = 0\}$.

Unique equilibrium prices at $\mathbf{p} = (\frac{1}{\sqrt{2}}, 0)$



Results

How hard is it to find a competitive equilibrium?

Theorem 1: Finding an exact competitive equilibrium in the circuit model is FIXP-complete.

Theorem 2: Finding an approximate equilibrium in the algorithm or query model takes a polynomial number of queries / steps if the market consists of two agents.

Theorem 3: Finding an approximate equilibrium in the algorithm model is PPAD-complete, even if we restrict the market to three identical agents.

Theorem 4: Finding an approximate equilibrium in the query model takes exponentially many query in the approximation parameter for four or more agents.

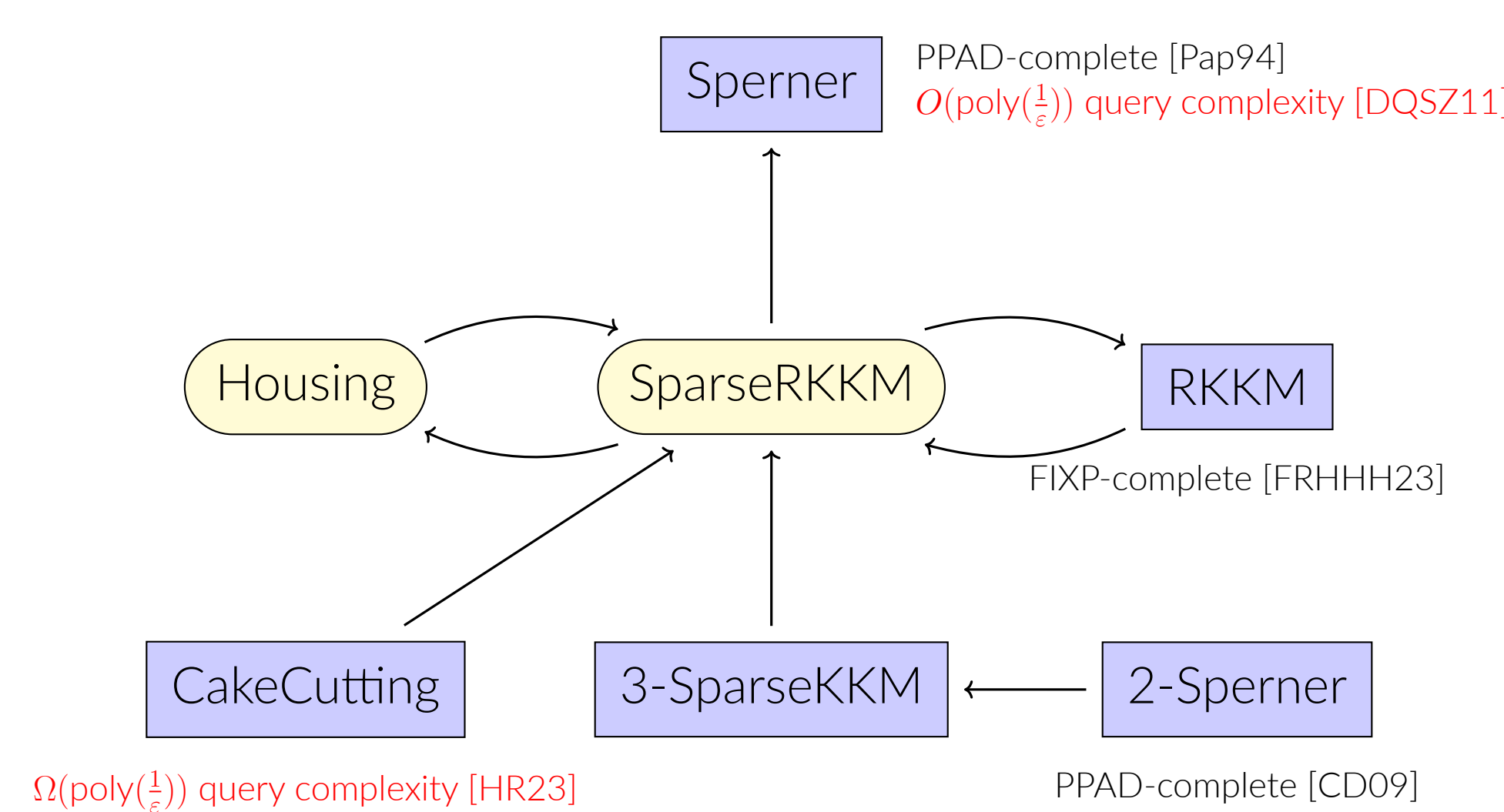
Implication: We can't find competitive equilibria efficiently.

Complexity Classes

PPAD: approximate Nash equilibrium, approximate market equilibria in Arrow-Debreu and Fisher markets, cake cutting, Sperner, End-of-the-Line.

FIXP: (Exact) Nash equilibrium, KKM, Rainbow-KKM, etc.

Proof Overview



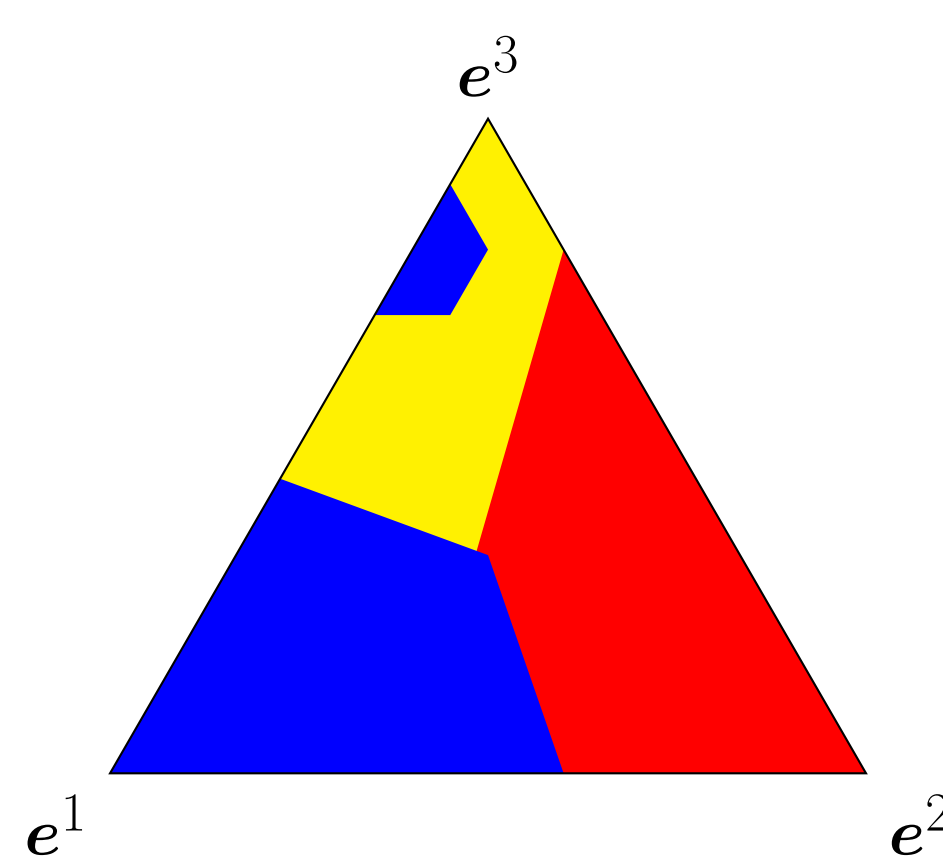
(Rainbow-)KKM

Standard simplex $\Delta_{n-1} := \text{conv}\{\mathbf{e}^j \mid j \in [n]\}$.

F_S is the face of Δ_{n-1} spanned by vertices $\{\mathbf{e}^j \mid j \in S\}$.

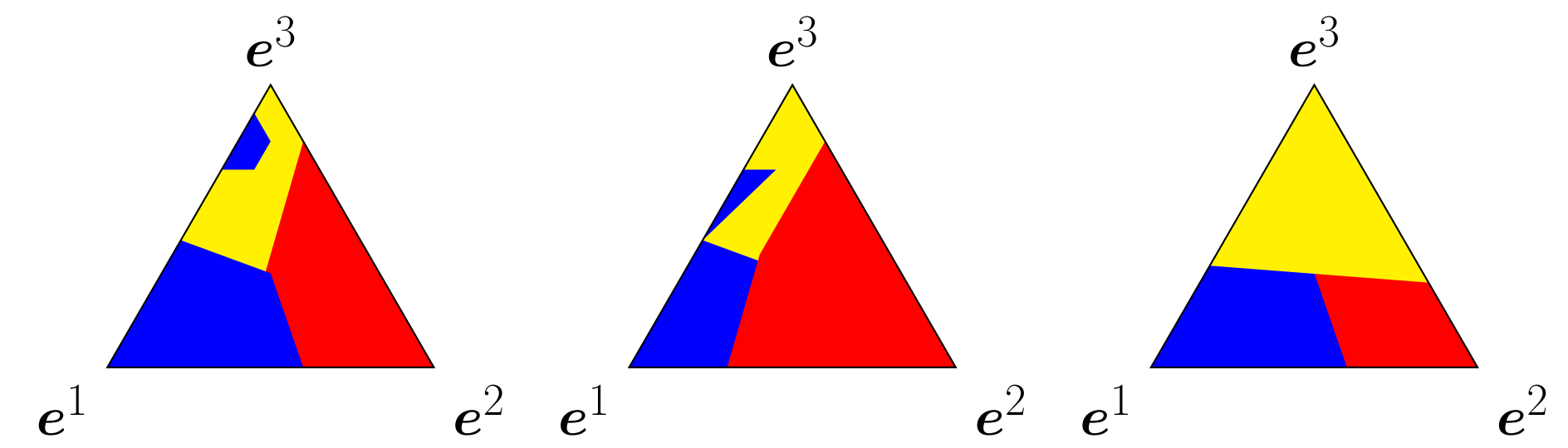
KKM covering: $C_1, \dots, C_n \subseteq \Delta_{n-1}$ s.t. $F_S \subseteq \bigcup_{j \in S} C_j$ for all $S \subseteq [n]$.

Sparse KKM covering: For every $j \in [n]$, $\mathbf{x} \in C_j$ implies $x_j > 0$.



(Approximate) KKM Lemma (cf. Knaster et al., 1929):

- For any KKM covering $C_1, \dots, C_n$ of closed sets, there exists $\mathbf{x}$ contained in every C_j .
- For any KKM covering $C_1, \dots, C_n$ and $\varepsilon > 0$, there exists $\mathbf{x}$ ε -close to each C_j .



(Approximate) Rainbow-KKM Lemma (cf. Gale, 1984):

- Any collection of n KKM coverings $C_1^i, \dots, C_n^i$ with closed sets admits a point $\mathbf{x}$ and permutation π of $[n]$ so that $\mathbf{x}$ lies in every $C_{\pi(i)}^i$.
- Any collection of n KKM coverings $C_1^i, \dots, C_n^i$, admits a point $\mathbf{x}$ and permutation π of $[n]$ so that $\mathbf{x}$ is ε -close to every $C_{\pi(i)}^i$.

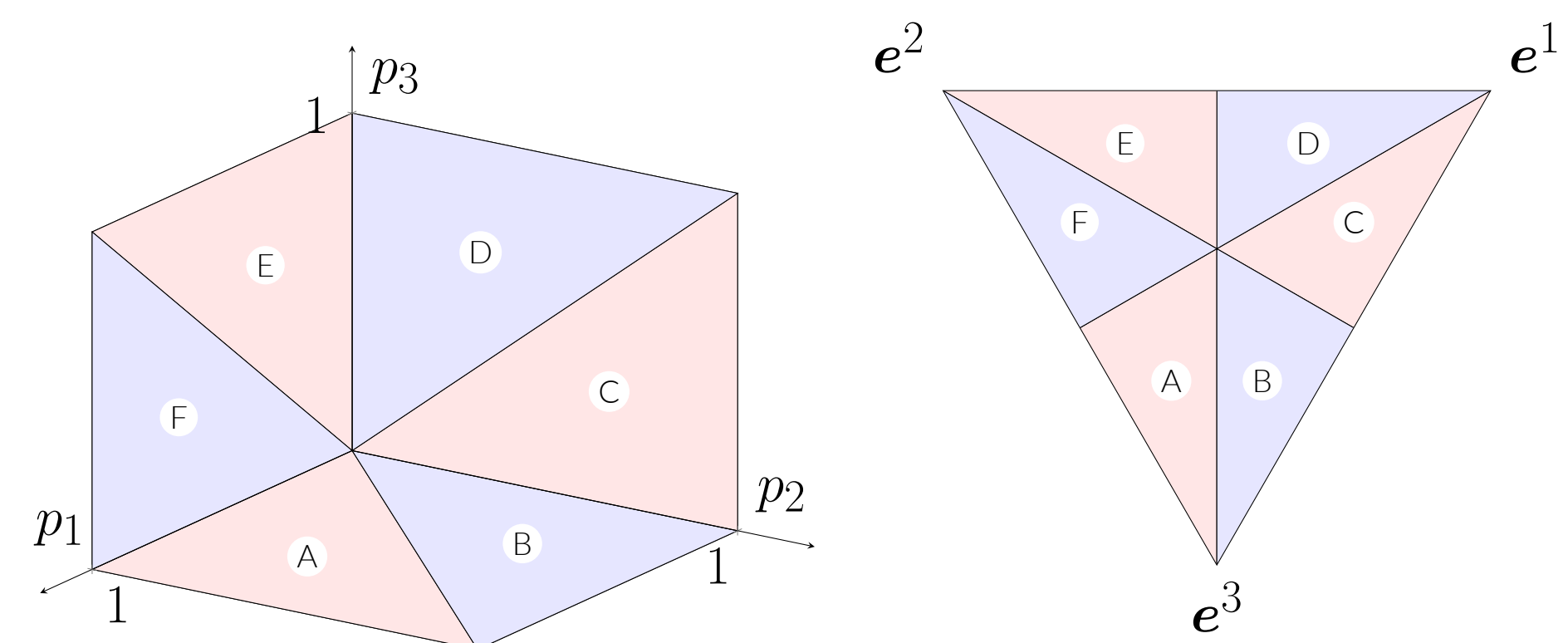
The Equivalence of Housing and Sparse RKKM

Gale (1984) defines **homeomorphism** ϕ .

Divide domain of housing market into $n!$ simplices corresponding to permutations π of $[n]$.

$$\Sigma_\pi := \{\mathbf{p} \in \Sigma_n \mid p_{\pi(1)} \geq p_{\pi(2)} \geq \dots \geq p_{\pi(n)} = 0\}$$

$$\phi(\mathbf{p})_{\pi(k)} := \frac{1 - p_{\pi(1)}}{n} + \frac{p_{\pi(1)} - p_{\pi(2)}}{n-1} + \frac{p_{\pi(k-1)} - p_{\pi(k)}}{n-k+1}.$$

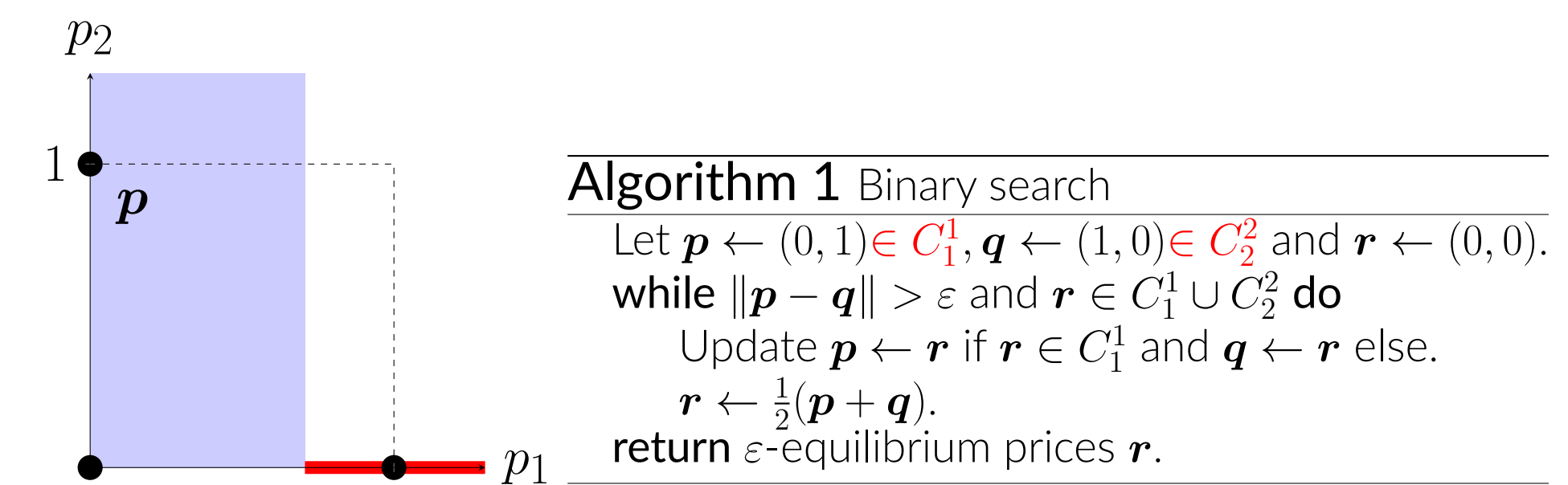


The Reductions

	Housing		SparseRKKM
Instance	$(P_1^i, \dots, P_n^i)_{i \in [n]}$	$\Rightarrow$	$(\phi(P_1^i), \dots, \phi(P_n^i))_{i \in [n]}$
Solution	$\phi^{-1}(\mathbf{x}), \pi$	$\Leftarrow$	$\mathbf{x}, \pi$

Lemma: ϕ is n -Lipschitz and ϕ^{-1} is n^2 -Lipschitz, which gives reductions for approximation versions.

Two-Agent Markets



Outlook

We initiate the study of housing markets from the perspective of computational complexity.

We hope these results will stimulate further examination of the complexity of markets with income effects:

- What is the complexity of Housing with utility functions instead of preference sets?
- Does hardness continue to hold under the natural assumptions of monotonicity in money and no externalities as in (Quinzii, 1984; Svensson, 1984)?
- How might CakeCutting with monotonic valuations (Deng et al., 2012; Hollender & Rubinstein, 2023) map to assumptions for Housing?

References

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